

# Assignment 6

## Part 1 — System Restatement (From Assignment 5)

Restate your system in a concise form:

- Decision statement

The decision hinges upon an algorithm that determines eligibility of a person for assisted suicide.

- Actor set

- 1) Data scientists, that are monitoring the inputs of data into the LLM system.
- 2) The System itself which is a statistical algorithm of neurons with predetermined by its training weights.
- 3) The doctors who will perform the operation in accordance to the conclusion of the machine.

- Primary and secondary targets

Primary targets are people submitting for assisted suicide.

Secondary targets are medical companies providing necessary medicine/equipment.

- Determinants and contributors

- Mental state of the patient.
- Physical state of the patient.
- \*Service providers (Service means euthanasia)
- Close ones of the person seeking \*service
- Algorithm making the decision (LLM)
- Key data/information inputs
- Medical information Source: Hospital/Medical databases.
- Personal information Source: Normalised questionnaire filled in by the patient.

## Part 2 — Mediation Pathway Inventory

- Data pathway(s)
- Patients medical data + application data (Local physical/virtual)-> Algorithms Database (remote Virtual, assuming the algorithm has centralised servers)

Here agency is limited to preservation of the data on the personal side by the patient/medical authority, and on the virtual side by

- Decision pathway(s)
- (Origin) Patient personal decision (will) (Personal) -> Patient request (Local Virtual) -> Algorithm analysis (remote Virtual) -> Algorithm decision (remote Virtual (perhaps physical if the decision is printed for physical archives) -> Person gets informed about the decision (Local Virtual), Medical authorities get informed (Remote virtual) (destination(s))

Physical agency is exercised by the Patient seeking \*Service, doctors providing the \*Service. Virtual agency is exercised by the algorithm itself.

- Accountability pathway(s)
- Accountability is in particular transferred from the decision integrity of the algorithm to the medical authority that adopts said algorithm. Assuming consent is set as a primary decider and can be withdrawn at any point of the process, also requiring consent of the patient AND the medical authority based on the decision of the algorithm, accountability primarily will lie on the person seeking \*Service.

Accountability pathway therefore will approximately look like this

Personal Consent (Personal)-> Signed Consent form (Local Virtual/Physical) -> Medical decision (Remote Virtual/Physical)

## Part 3 — Relativistic Physical and Virtual Travel Description

All of the above pathways were written from the POV of the

## Part 4 — Pathway Transformation and Friction Analysis

There is one primary information pathway being the pathway of medical data and the return pathway of decision and consent which can be assumed to be one cyclical transformative pathway.

- Where information or agency is transformed

When the person suits a request it is transformed from personal to virtual and the accountability from then on is put on the algorithm/medical authority as by the assumption the patient is trusting the algorithm with that decision, even if will is expressed from patient side.

- Where physical agency and virtual agency align, conflict, or become asymmetric

In this pathway agencies in every part of the pathway align. However where it may misalign is in the medical authority and the algorithm assuming a scenario where the decision of the algorithm is conflicting with the perception of the medical authority that adopted the algorithm.

- Where information access is asymmetric (one side can see/use/alter information while others cannot)

In an ideal scenario of non-malicious and competent agents, access to information is transparent from both patient and medical authority side. However the decision of the algorithm in case of a malicious actor may be altered if accessed and tampered with, as the information given into the program might go through medical authority first.

- Where pathway opacity increases

As the LLM is considered a black box algorithm as it is impossible to visualise the weights used in the equations that result in the final outcome, this is where opacity is at its worst.

- Where authority and responsibility diverge

When Authority gives a decision the responsibility then is out fully on the patient to maintain or withdraw consent.

- Where ethical distance increases

Decision of the medical authority for or against the \*Service is where the ethical distance is the highest. As potentially the patient

has most to lose, even with their own consent, while the medical authority (doctor/algorithm) risk nothing.

- Where feedback loops are present or absent

The interplay between patient request and algorithm/medical authority decision is in itself a feedback loop. Although false positives are nearly impossible to evaluate due to the nature of Euthanasia.

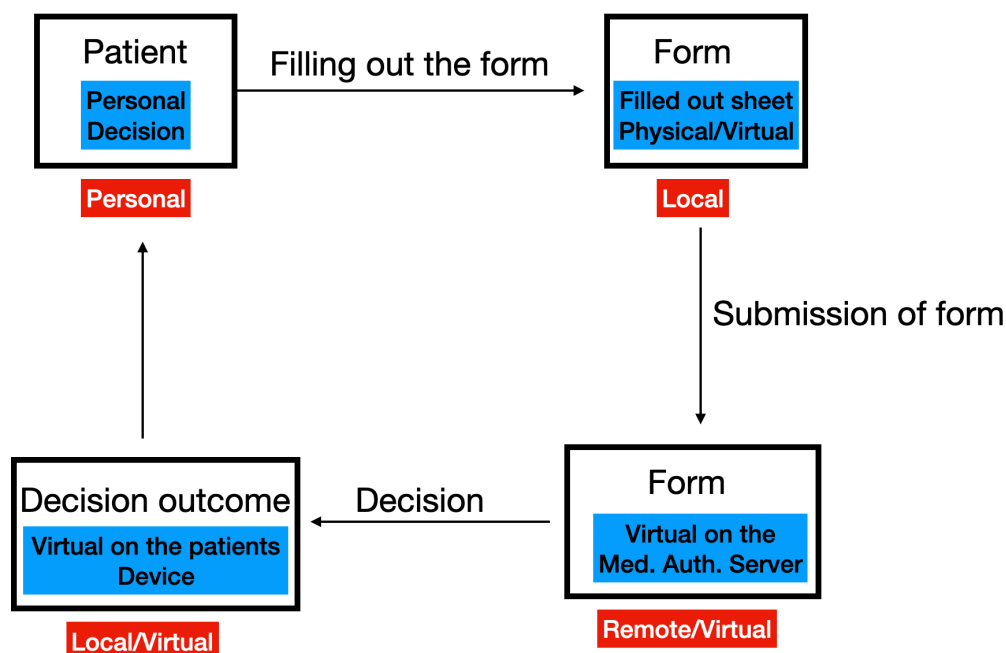
- Where the assumed symmetry between **cognitive intelligence** and **computational intelligence** holds, and where it breaks

In the case of the program having the authority of the evaluation outcome.

- Where the AI system is treated as a **mere tool** versus where it functions more like a **collaborative partner** in the pathway

In this pathway in an ideal scenario the AI system is treated as a tool that gives out an unbiased decision. However due to the nature of it being black box, true error rates would be hard to evaluate.

## Part 5 — Visual/Schematic Representation



Caption

